SRASHTI GUPTA

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Summary

Software Engineer with 5+ years of experience building Java/Spring Boot microservices, AWS cloud platforms, and full-stack systems. MS CS candidate specializing in AI-powered applications using Python, LangChain, and multi-agent workflows.

Education

Master of Science, Computer Science **GPA: 4.0**
Northeastern University, Boston, MA *Expected: May 2027*
Coursework: Algorithms, Computer Systems, Database Management Systems, Programming Design Paradigm

Bachelor of Technology, Information Technology **GPA: 3.7**
Banasthali Vidyapith, Rajasthan, India *May 2019*
Coursework: Data Structures, Theory of Computation, OOP, Software Engineering, Data Mining, Artificial Intelligence

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, Rust, C++, SQL, R, Shell Scripting
AI Engineering: LangChain, LangGraph, LLM APIs, AWS Bedrock, Claude, Groq, Multi-Agent Orchestration
Frameworks: Spring Boot, Spring Data JPA, Hibernate, Node.js, Express, FastAPI, React, Next.js, Redux
Databases & Cloud: AWS (Lambda, Step Functions, S3, EC2, RDS, SQS, Bedrock, CloudFront, Cognito), PostgreSQL, MySQL, DynamoDB, MongoDB, Redis
DevOps & Testing: Git, Docker, Kubernetes, Apache Kafka, CI/CD, Jenkins, Swagger/OpenAPI, Maven, Gradle, JUnit, pytest

Projects

AlphaSign - Multi-Agent Financial Intelligence | *Python, Next.js* | Live Demo | Team Repository June 2026
- Built a multi-agent financial research backend for a hackathon using **LangGraph** and **LangChain**, enabling concurrent ticker-aware analysis and streaming live insight updates to the frontend with Server-Sent Events (SSE).

Autopsy Lab - AI Incident Reconstruction | *Python, FastAPI, Groq, Next.js* | Live Demo | GitHub June 2026
- Built a Madison AI Hackathon log-analysis tool that converts raw system dumps into root-cause timelines, severity, and JSON reports using Groq LLMs, **FastAPI**, and Pydantic validation.

Sage - AI Research Synthesizer | *Java, AWS Lambda, Bedrock, React, DynamoDB* | Live Demo | GitHub May 2026
- Built a serverless research synthesizer that generates structured reports in under 45 seconds by parallelizing AI research through a 5-state **AWS Step Functions** workflow using **Amazon Bedrock** Claude.

Adaptive Concurrent Web Server | *Rust, Python* | Team Repository | Case Study March 2026
- Engineered a multithreaded Rust server with an adaptive 4-16 worker pool scaling via an asynchronous lock-free MPSC channel architecture, eliminating P99 spikes by 40% (1.73ms to 1.04ms) under high load at 31,725 req/s.

Streaming Analytics Data Warehouse | *R, MySQL, ETL* | GitHub December 2025
- Designed a year-partitioned MySQL star schema and containerized **R** ETL script parsing 98,472 records; optimized analytical join queries to expose a 34% Q4 audience viewing density peak over historical data.

Experience

Senior Software Engineer | **NAB Global Innovation Center, Gurgaon, India** February 2023 - July 2025
- Optimized 20+ **Java**/**Spring Boot** banking APIs on AWS and Kubernetes using OAuth 2.0, Redis caching, and database indexing, reducing latency 35% from 180ms to 117ms; earned STAR Award.
- Led a 4-month migration of 20 microservices from Java 8 to 17, resolving 8 critical CVEs by updating Spring Security protocols, and integrated **Kafka** (10K msg/s) with MongoDB via blue-green zero-downtime rollouts.
- Built a React and **REST API** tracking tool ingesting Jenkins metadata into **PostgreSQL** via automated HTTP polling, increasing engineering delivery visibility from 23% to 98%; earned SPOT Award.

Software Engineer | **IBM, Gurgaon, India** April 2021 - December 2022
- Architected Spring Data JPA and Hibernate microservices across an 18-table PostgreSQL schema, applying Circuit Breaker and Saga patterns to reduce P99 response time 45% from 420ms to 231ms.
- Designed 12-stage automated Jenkins **CI**/**CD** pipelines with Docker across 10+ services, decreasing release cycle times by 40% (65 to 39 minutes) while standardizing REST endpoints with Swagger/OpenAPI.
- Shipped full-stack features using **React** and Spring Boot alongside cross-functional partners, driving a 40% growth in Daily Active Users (12.3K to 17.2K) and extending average session lengths from 4.2 to 5.9 minutes.

Software Engineer | **Capgemini Engineering, Gurgaon, India** January 2019 - March 2021
- Engineered core data layers for Samsung and HughesNet backends using **Java**, IBM DB2, and MySQL with HikariCP connection pooling; optimized complex query patterns to secure 30% faster data retrievals.
- Resolved legacy N+1 database blocks over 14 distinct microservices reducing P95 response times 25% (520ms to 390ms) while maintaining a strict 98% TestNG/Mockito automated test coverage rate.